

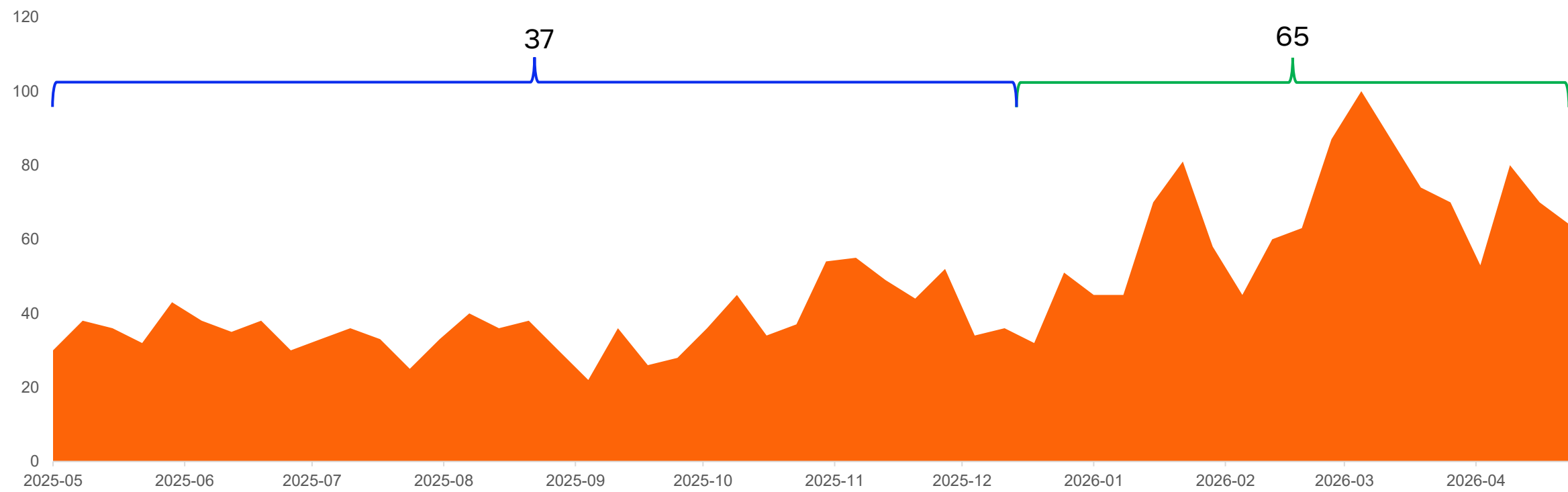


Electric Vehicles in South Africa

Evidence from the frontline...

Leading indicators are telling an electric car story

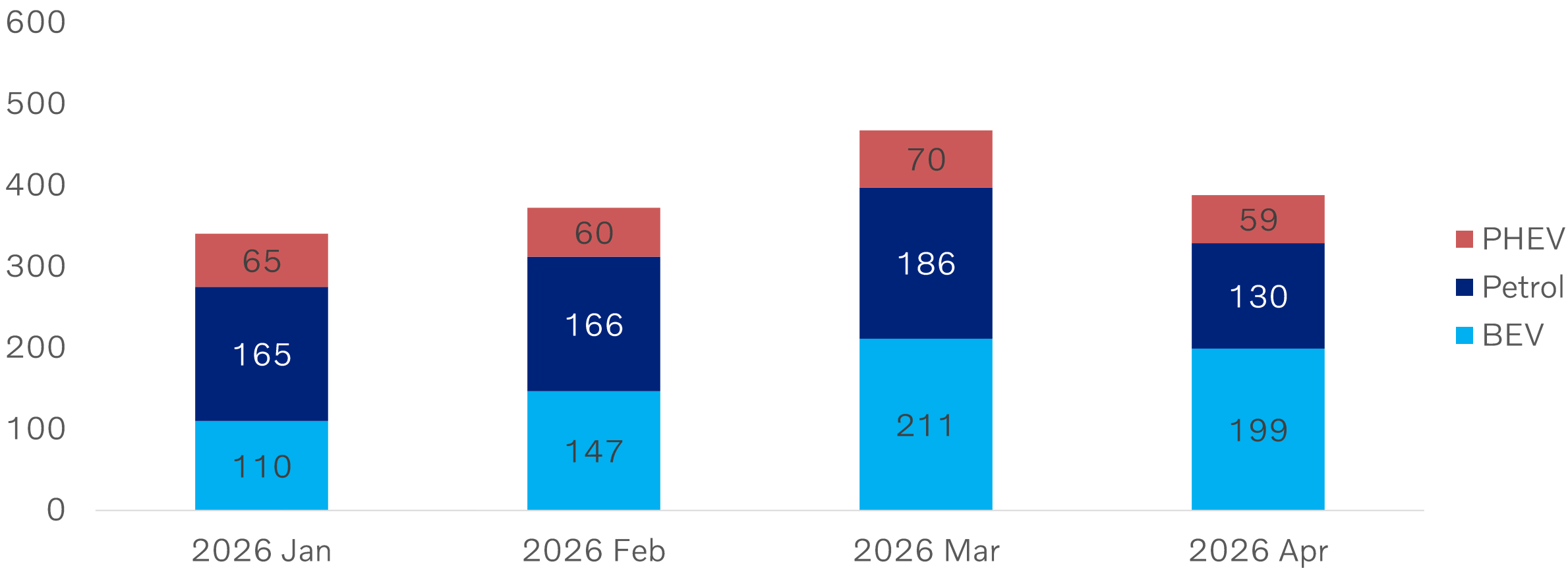
Google Keyword Search for Electric Cars and associated keywords in South Africa from May 2025 (weekly)



Source: Google, Keywords associated with Electric Cars, over 1 year.

Our own traffic tells an interesting story – although this is OEM specific

Average product sessions on Volvocars.com/za website by month



Source: Volvocars.com/za – customer traffic is averaged per day for the month

Running costs – a case study: Volvo EX30

Measured Time period & Energy:

- Charging Period: 2024/03/19 to 2026/03/28 (2 years)
- Charging Sessions logged: 188
- Total energy recorded: 4,991.01 kWh
- Total distance recorded: 28,744kms

Average sessions + Total Energy splits:

- Average session charge size: 26.55 kWh
- Average Monthly charging session: 207.9 kWh
- Paid sessions: 9
- "Free/Solar" sessions: 179 (home charging via excess Solar + workplace charging)
- Paid Energy: 141.29 kWh
- "Free energy": 4,849.72 kWh

AC vs DC Charge sessions:

- DC Charge sessions: 5 | 96.81kWh | R 696.86
- Non-DC Charge Sessions: 183 | 4,893.20 kWh | R 273.22
- Total Cost for 2 years: R970.08

Cost/Km:

- Average cost / km: **R 0.04/km** for an EV charging at home/work*
- Typical cost / km: **R 1.35/km** for an EV charging on public DC**

* Assuming 90%+ charging off excess home solar or at a workplace

** Assuming ave. DC charge rate R 9.00/kWh, ave. EV consumption 15kWh/100kms

Petrol Equivalent calculations

Total Distance Driven:

- 28,744kms
- Assume 7.5 litres/100km consumption
- Total litres of fuel required: 2,155 (28,744 x 0.075)
- Estimated petrol spend: R 48,277 (2155 x R22.40/l)
- Less EX30 charging costs: R 48,277 – R970.08 = R 47,307

Cost/Km:

- Average cost / km: **R1.68/km** for petrol (@ R22.40/l)

Therefore ~R 47,307 in fuel costs avoided over a 2-year period

Emissions Offset:

- CO2 avoided by EV: 3,777 kg CO2 (across 2 years)
- **Percentage reduction vs petrol: ~ 55% lower**
- Petrol emissions intensity: 237.5g/km
- EV Blended Intensity: ~106 g/km

Using the Volvo EX30's real-world efficiency and adjusting petrol emissions for fuel extraction and refining, it delivers **~55% lower operational CO₂ emissions than an equivalent petrol vehicle**, with home and workplace solar providing a material advantage.

“Today, electric cars often have a lower total cost of ownership than ICE cars over the vehicle lifetime, due to reduced fuel and maintenance expenses.”

(International Energy Agency) IEA Global EV Outlook 2025

..... On average, the overall cost of operating a BEV consistently outperforms that of a comparable ICE. In fact, across the countries examined in this study, several scenarios strongly indicate that BEVs present a more cost-effective option

Examining total cost of ownership: electric vs internal combustion engine vehicles, Fabio Carello Ayvens (2024)

But, is this actually true in South Africa?

Purchase Price

EVs more expensive relative to ICE

Operating costs

Lower running and maintenance (except tyres / insurance?)

Residual value

To early to tell

Public charging is becoming faster and more accessible too....

- **Coverage:** Major highways (N1, N2, N3) have charge points spaced within ~300 km. Urban EV drivers in Johannesburg, Cape Town and Durban can locate a charge point within 20 km.
- **Charging sessions:** Rubicon reports that total charging sessions doubled in 2024 versus 2023, reflecting the growth in EVs on South African roads. Early indications for 2026 show charging up 64%

VOLVO ES90

800V DC CHARGE TEST SESSION @ ZERO CARBON CHARGE

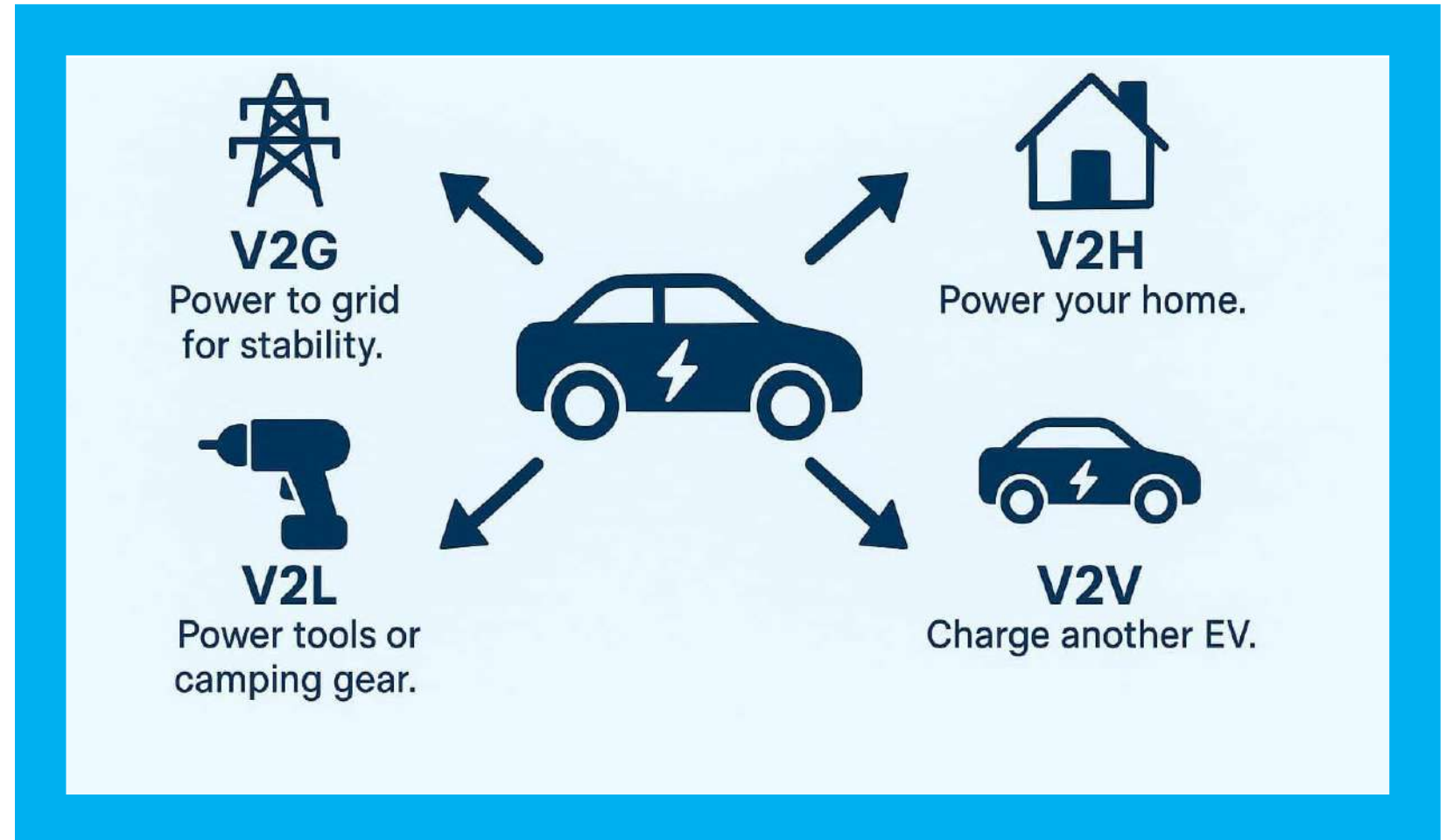
Wednesday 21 May 2026

Three Towers Charge Station (N3)



But, here is something coming in EVs that could change everything.....

Bi-directional charging, everything from your appliances, to home, to other EVs and to feedback into the grid



Lastly...

1. Africa's Motorisation Gap

Africa has the world's lowest vehicle penetration rate. As GDP per capita rises across Sub-Saharan Africa, **vehicle demand will grow** substantially.

South Africa is uniquely positioned — with established manufacturing, a port network, and AfCFTA market access — to serve this demand with locally assembled NEVs.

3.A Business Model That Works: Local + Export

A **commercially viable** NEV assembly operation requires **sufficient domestic offtake AND export volume**.

2. The Used Vehicle Risk

Growing imports of used ICE vehicles into Sub-Saharan Africa are displacing demand for new vehicles, undermining fleet modernisation and complicating the region's decarbonisation pathway.